Narrower Is Not Automatically Better

AP Statistics · Unit 3 · Topic 3.3 · B: 2026-11-30 · E: 2027-01-04 · TEACHER KEY

CED TETHER

- **Skill 1.D** — Identify an appropriate inference method for confidence intervals.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **Skill 3.D** — Construct a confidence interval, provided conditions for inference are met.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.A** — Identify an appropriate confidence interval procedure for a population proportion. [Skill 1.D]
- **EK UNC-4.A.1** — The appropriate confidence interval procedure for a one-sample proportion for one categorical variable is a one sample z-interval for a proportion.
- **LO UNC-4.B** — Verify the conditions for calculating confidence intervals for a population proportion. [Skill 4.C]
- **EK UNC-4.B.1** — In order to make assumptions necessary for inference on population proportions, means, and slopes, we must check for independence in data collection methods and for selection of the appropriate sampling distribution.
- **EK UNC-4.B.2** — In order to calculate a confidence interval to estimate a population proportion p, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: (i) Data should be collected using a random sample or a randomized experiment. (ii) When sampling without replacement, check that n $\leq$ 10%N, where N is the size of the population.
- b. To check that the sampling distribution of p-hat is approximately normal (shape): for categorical variables, check that both the number of successes, n·p-hat, and the number of failures, n(1 - p-hat) are at least 10.
- **LO UNC-4.C** — Determine the margin of error for a given sample size and an estimate for the sample size that will result in a given margin of error for a population proportion. [Skill 3.D]
- **EK UNC-4.C.1** — The standard error of p-hat is SE = sqrt(p-hat(1 - p-hat)/n).
- **EK UNC-4.C.2** — A margin of error gives how much a value of a sample statistic is likely to vary from the value of the corresponding population parameter.
- **EK UNC-4.C.3** — For categorical variables, the margin of error is z* $\times$ sqrt(p-hat(1 - p-hat)/n) for a one-sample proportion.
- **EK UNC-4.C.4** — The formula for margin of error can be rearranged to solve for n, the minimum sample size needed to achieve a given margin of error. Use a guess for p-hat or use p-hat = 0.5 to find an upper bound for the sample size.
- **LO UNC-4.D** — Calculate an appropriate confidence interval for a population proportion. [Skill 3.D]
- **EK UNC-4.D.1** — For a one-sample proportion, the interval estimate is p-hat $\pm$ z* $\times$ sqrt(p-hat(1 - p-hat)/n).
- **EK UNC-4.D.2** — Critical values represent the boundaries encompassing the middle C% of the standard normal distribution, where C% is an approximate confidence level for a proportion.
- **LO UNC-4.E** — Calculate an interval estimate based on a confidence interval for a population proportion. [Skill 3.D]
- **EK UNC-4.E.1** — Confidence intervals for population proportions can be used to calculate interval estimates with specified units.

Essential question: Why can a wider interval from a random sample be more trustworthy than a narrower formula result?

DOK-3 skill: 4.C · **FRQ pattern:** `narrower-self-selected-interval-vs-wider-random-interval`

DOK rationale: Part (c) requires judging two competing interval reports by coordinating recruitment, numerical conditions, and interval width, then explaining why the more precise-looking output cannot support the target-population claim.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in)DOK: 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to choose a plan before calculating.
- Begin the 6.1–6.2 video at minute 5; return at minute 33 to separate calculation from generalization.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose a plan and cite one collection detail.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

Questions to ask

- Which condition depends on how cardholders entered the sample?
- What do the numerical conditions not establish?
- Why is Plan B's formula interval narrower?

Adult role
Solo teacher. Circulate 5 pairs. Require all three checks and distinguish a formula output from a defensible interval.

[WARN] WATCH FOR

- Treating a larger self-selected group as representative.
- Using numerical checks as substitutes for random selection.
- Calling Plan B's output a population confidence interval.

SCORING PART (c) — E / P / I

Expected elements

- **selects-plan-a** (required) — Chooses Plan A for population inference
- **uses-collection-and-numerical-evidence** (required) — Cites the SRS and one numerical check
- **contrasts-widths-with-validity** (required) — Notes $0.066 < 0.137$ but Plan B is nonrandom
- **states-reader-consequence** (required) — Explains false precision from self-selection

E	Chooses A, cites both kinds of checks, contrasts widths, and explains self-selection.
P	Chooses A but omits a check, width, or reader consequence.
I	Chooses B for size or width, or treats numerical checks as randomness.

[STAR] TODAY'S PROBLEM — annotated

A library system has 20,000 adult cardholders and wants to estimate p, the proportion who used a digital audiobook last month.

Plan A: A computer takes an SRS of 200 cardholder IDs. All answer; 116 say yes.

Plan B: A newsletter link lets cardholders choose to respond once. Of 800 respondents, 520 say yes.

Eli: "Use Plan B. Its larger sample gives the narrower interval."

Dana: "Use Plan A. Entry into the sample matters more than the narrowest output."

Plan counts

Plan	N	n	Yes	No
A	20,000	200	116	84
B	20,000	800	520	280

FIRST TAKE (5 min)

Which plan, if either, should estimate all adult cardholders? Cite one collection detail.
Key: Not graded. The larger sample is intentionally attractive.

Rules callout "ONE-SAMPLE INTERVAL FOR A PROPORTION (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define the population parameter p in context and name the confidence-interval procedure that would be appropriate when its conditions are met.

Key: Here p is the proportion of all 20,000 adult library cardholders who used a digital audiobook last month. When conditions are met, the procedure is a one-sample z-interval for a population proportion.

DOK 2 **(b)** For each plan, compute $\hat{p}$, check the random, 10 percent, and large-count conditions, and calculate the 95 percent formula interval. Mark any nondefensible population interval.

Key: Plan A has $\hat{p} = 116/200 = 0.58$. It is an SRS; $200 \leq 0.10(20{,}000) = 2{,}000$; and counts are 116 and 84. Thus $SE = 0.0348999$, $ME = 1.96(SE) = 0.0684037$, and the valid interval is $(0.5116, 0.6484)$. Plan B has $\hat{p} = 520/800 = 0.65$; $800 \leq 2{,}000$ and counts 520 and 280 pass, but self-selection fails the random condition. Its mechanical output is 0.65 ± 0.0330523, or $(0.6169, 0.6831)$, not a defensible population confidence interval.

DOK 3 **(c)** Decide which plan can support a 95 percent interval for all cardholders. Cite recruitment and a numerical condition, compare interval widths, and explain the reader consequence of reporting Plan B's output.

Key: Use Plan A. Its SRS, $200 \leq 2{,}000$, and counts 116 and 84 support the interval $(0.512, 0.648)$ for all cardholders. Its width is about 0.1368 versus Plan B's smaller formula width, 0.0661. Plan B's volunteers may differ systematically from nonrespondents; its margin of error covers random variation, not self-selection. Reporting it would falsely suggest that a narrow range represents all cardholders with 95 percent confidence.

EXIT

One thing the video changed about my first take: ______________________________